Contents

1 Topological Evidence Curves for Anytime-Valid Predictive Maintenance **1**
1.1 Abstract . . . 1
1.2 1. Problem Setup . . . 1
1.3 2. Topological Evidence Features . . . 1
1.4 3. Main Guarantees . . . 2
1.4.1 Theorem 1 (Anytime Validity, Inversion Form) . . . 2
1.4.2 Theorem 2 (Width Contraction Under Separation) . . . 3
1.4.3 Theorem 3 (Topological Phase Separation) . . . 3
1.4.4 Theorem 4 (Stability) . . . 4
1.5 4. Algorithm . . . 4
1.6 5. Experimental Protocol . . . 4
1.6.1 Synthetic Validation . . . 4
1.6.2 C-MAPSS Stress Test . . . 4
1.7 6. Implementation Notes (This Repository) . . . 5
1.8 7. Limitations and Scope . . . 5
1.9 8. Reproducibility . . . 5
1.10 9. Canonical Result Snapshot (2026-03-10) . . . 5
1.11 10. Auto-Compiled Figures (Canonical Build) . . . 6
1.11.1 External Metrics Table . . . 6
1.11.2 Figures . . . 6

1 Topological Evidence Curves for Anytime-Valid Predictive Maintenance

1.1 Abstract

We introduce **Topological Evidence Monitoring (TEM)**, a sequential predictive-maintenance method that combines conformal e-value accumulation with topological summaries of hypothesis-indexed evidence curves. Instead of emitting a single alarm statistic, TEM maintains a full evidence profile over candidate remaining useful life (RUL) values and tracks the evolution of that profile via 0-dimensional persistence. The method provides (i) an anytime-style confidence set over RUL hypotheses, (ii) a topological separation signal through persistence-gap growth, and (iii) efficient deployment on GPU-based RUL models with low CPU overhead for evidence updates.

1.2 1. Problem Setup

At monitoring step t, a base model outputs $\hat{R}_t$ for unknown true RUL R_t. Let $\mathcal{A}^{cal} = \{A_i^{cal}\}_{i=1}^{n_{cal}}$ be calibration residuals, typically $A_i^{cal} = |\hat{R}_i - R_i|$ from held-out healthy/validation data.

For a candidate RUL value $r \in \{1, \dots, R_{max}\}$, define:

$$s_t(r) = |\hat{R}_t - r|, \quad p_t(r) = \frac{1 + \#\{i : A_i^{cal} \geq s_t(r)\}}{n_{cal} + 1}.$$

With betting parameter $\lambda_t \in [0, 1]$:

$$K_t(r) = K_{t-1}(r) \cdot (1 + \lambda_t(1 - 2p_t(r))), \quad K_0(r) = 1.$$

Define the log-evidence curve:

$$V_t(r) = \log K_t(r).$$

The anytime-style confidence set is:

$$C_t^\alpha = \{r : K_t(r) < 1/\alpha\}.$$

1.3 2. Topological Evidence Features

Treat $V_t(\cdot)$ as a scalar function on a 1D grid and compute H_0 persistence on its sublevel filtration.

Key summaries: - π_t: largest persistence lifetime (deepest valley) - $\pi_t^{(2)}$: second-largest persistence - $\gamma_t = \pi_t/\pi_t^{(2)}$: persistence-gap ratio - $r_t^* = \arg\min_r V_t(r)$: most-evidenced RUL - $W_t^\alpha = |C_t^\alpha|$: confidence width

Decision policy: - **ALERT** if $\gamma_t > \gamma_{crit}$ and $W_t^\alpha < W_{crit}$ - **CRITICAL** if no plausible RUL remains over sustained steps

1.4 3. Main Guarantees

Let $(\mathcal{F}_t)_{t \geq 0}$ be the monitoring filtration. In the implementation, evidence is maintained on a fixed hypothesis grid (failure-time index τ_h); the RUL curve is obtained by re-indexing $V_t(r) = \log K_t(t + r - 1)$. Guarantees are stated on the fixed grid and transferred to the derived RUL set.

Assumptions used below: - **A1 (Conditional super-uniformity at truth):** for the true fixed hypothesis $h^\star$, $p_t(h^\star)$ satisfies

$$\Pr(p_t(h^\star) \leq u \mid \mathcal{F}_{t-1}) \leq u, \quad \forall u \in [0, 1].$$

- **A2 (Predictable bounded betting):** $\lambda_t \in [0, 1]$ is $\mathcal{F}_{t-1}$-measurable. - **A3 (Wrong-hypothesis drift):** for each wrong hypothesis $h \neq h^\star$, the log increment

$$X_t(h) = \log\big(1 + \lambda_t(1 - 2p_t(h))\big)$$

has conditional mean at least $\rho_h > 0$ after a finite transient. - **A4 (Increment tails):** centered increments are conditionally sub-Gaussian with proxy variance σ_h^2.

1.4.1 Theorem 1 (Anytime Validity, Inversion Form)

Under A1-A2, for the true hypothesis $h^\star$, $K_t(h^\star)$ is a nonnegative supermartingale with $K_0(h^\star) = 1$. Therefore,

$$\Pr\left(\sup_{t\geq 0} K_t(h^\star) \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Equivalently,

$$\Pr(\exists t : h^\star \notin C_t^\alpha) \leq \alpha.$$

Proof. Define the multiplicative factor

$$M_t(h^\star) = 1 + \lambda_t(1 - 2p_t(h^\star)).$$

By A2, M_t is $\mathcal{F}_{t-1}$-measurable except for p_t. By A1,

$$\mathbb{E}[1 - 2p_t(h^\star) \mid \mathcal{F}_{t-1}] \leq 0,$$

hence $\mathbb{E}[M_t(h^\star) \mid \mathcal{F}_{t-1}] \leq 1$. Since $K_t = K_{t-1}M_t$,

$$\mathbb{E}[K_t(h^\star) \mid \mathcal{F}_{t-1}] \leq K_{t-1}(h^\star),$$

so $K_t(h^\star)$ is a nonnegative supermartingale. Ville's inequality gives

$$\Pr\left(\sup_{t} K_t(h^\star) \geq 1/\alpha\right) \leq \alpha.$$

The confidence set is $C_t^\alpha = \{h : K_t(h) < 1/\alpha\}$, giving the stated inversion result. □

1.4.2 Theorem 2 (Width Contraction Under Separation)

Under A2-A4 and A3 for all wrong hypotheses, each wrong $h \neq h^\star$ crosses the rejection boundary in finite time with high probability, and

$$\tau_h(\alpha, \delta) = O\left(\frac{\log(1/\alpha) + \log(1/\delta)}{\rho_h}\right).$$

Consequently, W_t^α contracts as t grows.

Proof. Let $S_t(h) = \log K_t(h) = \sum_{s=1}^t X_s(h)$. By A3,

$$\mathbb{E}[X_s(h) \mid \mathcal{F}_{s-1}] \geq \rho_h.$$

By A4 and standard sub-Gaussian concentration for martingale differences, with probability at least $1 - \delta$,

$$S_t(h) \geq t\rho_h - \sigma_h\sqrt{2t\log(1/\delta)}.$$

Hence $S_t(h) \geq \log(1/\alpha)$ once t exceeds the stated order, so $h \notin C_t^\alpha$. Applying this per wrong hypothesis and union-bounding over $h \neq h^\star$ yields contraction of W_t^α. □

1.4.3 Theorem 3 (Topological Phase Separation)

Assume post-onset drift induces monotone separation between the true valley and dominant wrong-hypothesis valleys, while perturbations obey A4 uniformly over hypotheses. Then the dominant persistence gap obeys

$$\gamma_t = \frac{\pi_t}{\pi_t^{(2)}} \to \infty$$

in probability after onset, with detection delay scaling

$$O\left(\frac{\sigma^2 \log(R_{max}/\delta)}{\rho^2}\right),$$

where ρ is the minimal post-onset drift gap.

Proof. Write evidence as $V_t(r) = m_t(r) + \xi_t(r)$, where m_t is the drift part and ξ_t is stochastic fluctuation. Under the drift condition, the depth difference between the true basin and the best competing basin in m_t grows at least linearly in elapsed degraded time, $c_1\rho t$. By A4 and union bounds over $r \in \{1, \dots, R_{max}\}$, the sup fluctuation of ξ_t is $O(\sigma\sqrt{t\log(R_{max}/\delta)})$. Persistence lifetimes are 1-Lipschitz in sup norm (via diagram stability), so observed basin-depth differences inherit the same signal-minus-noise form. Linear signal dominates square-root noise after t of the stated order, implying persistent growth of γ_t. □

1.4.4 Theorem 4 (Stability)

For two evidence curves V_t, V_t', let PD_t, PD_t' be their sublevel persistence diagrams. Then

$$d_B(PD_t, PD_t') \le \|V_t - V_t'\|_\infty.$$

Therefore any Lipschitz functional of PD_t (e.g., top persistence, gap ratio after denominator clipping, count above threshold) is stable to bounded predictor perturbations.

Proof. The bottleneck stability inequality for sublevel-set persistence on tame functions gives the first bound directly. Composing with Lipschitz statistics preserves continuity with at most multiplicative Lipschitz constants. □

1.5 4. Algorithm

At each monitoring step: 1. Predict $\hat{R}_t$ 2. Compute $p_t(r)$ for all $r \in [1, R_{max}]$ 3. Update $K_t(r)$, obtain $V_t(r)$ 4. Compute persistence summaries $(\pi_t, \pi_t^{(2)}, \gamma_t)$, r_t^*, W_t^α 5. Trigger policy based on (γ_t, W_t^α)

Complexity per step: - Evidence update: $O(R_{max} \log n_{cal})$ with sorted calibration residuals - 1D persistence: $O(R_{max} \log R_{max})$ (or near-linear with specialized union-find)

Implementation note: in code we maintain a fixed hypothesis grid over τ_h and derive the current-RUL evidence curve by $V_t(r) = \log K_t(t + r - 1)$. This preserves the fixed-parameter structure needed for theorem-aligned τ-coverage diagnostics.

1.6 5. Experimental Protocol

1.6.1 Synthetic Validation

- Simulate engines with known onset and controlled drift/noise
- Check empirical anytime violations vs α
- Validate phase-separation behavior of γ_t

1.6.2 C-MAPSS Stress Test

- Data via `rul-datasets` with leakage-free train/val/test usage
- Train fast Conv1D baseline on RTX 4050 with AMP; use `torch.compile` when supported, otherwise run eager mode
- Build calibration residuals from validation trajectories
- Run TEM on full test trajectories (`test` loaded with `alias="dev"`)
- Report alert rate, first-alert timing, temporal coverage, and trajectory plots

1.7 6. Implementation Notes (This Repository)

- `scripts/download_cmapss.py`: dataset preparation
- `scripts/train_fast_cmapss.py`: high-throughput training + checkpoint + calibration export
- `scripts/run_tem_cmapss.py`: TEM inference + plotting + metrics
- `scripts/run_synthetic_validation.py`: theorem stress checks
- `tem/topology.py`: fast 1D H_0 persistence
- `tem/evidence.py`: vectorized evidence curve updates

1.8 7. Limitations and Scope

- The formal guarantee requires calibration/test compatibility assumptions.
- 1D H_0 persistence is intentionally lightweight; extensions to 2D evidence surfaces can capture richer topology.
- C-MAPSS is a benchmark stress test, not proof of universal field validity.

1.9 8. Reproducibility

Set seeds, save checkpoints + calibration arrays, and archive all JSON metrics and PNG figures in `outputs/`.

1.10 9. Canonical Result Snapshot (2026-03-10)

This section pins the paper to the current canonical artifacts: - `outputs/external_performance_report.json` - `outputs/external_dataset_summary.json` - `outputs/publication_full_rtx4050/stats_conference_re`

External datasets (FD001): - FEMTO: RMSE=14.362, MAE=7.993, RMSE_last=52.873, MAE_last=50.395, RUL coverage=1.000, tau violation=0.000 - XJTU-SY: RMSE=30.133, MAE=23.684, RMSE_last=33.171, MAE_last=33.156, RUL coverage=1.000, tau violation=0.000 - C-MAPSS: RMSE=16.396, MAE=10.992, RMSE_last=16.623, MAE_last=12.219, RUL coverage=1.000, tau violation=0.000

Canonical external run settings: - `alpha=0.001`, `lambda_bet=0.03`, `pvalue_safety_margin=0.3` - Dataset-specific training overrides: - FEMTO: `epochs=15`, `low_rul_loss_weight=4.0`, `low_rul_threshold=25`, `low_rul_weight_power=1.0` - XJTU-SY: `epochs=15`, `low_rul_loss_weight=2.0`, `low_rul_threshold=20`, `low_rul_weight_power=1.5` - C-MAPSS: default training profile (`epochs=15`, no low-RUL reweighting override)

Readiness gate summary: - Legacy score is 10.0/10.0 (target 9+ passes), and conservative score is 10.0/10.0 (target 9+ passes). - FD004 tau-identifiability now passes strict per-FD threshold after the promoted `max_rul=130` run (`187/248 = 0.7540 >= 0.75`), with pooled ratio `0.7963`. - Strict regime deep check is clean across FD001-FD004 under the stricter finite-sample setting (`num_findings_total=0`).

Interpretation caveat: - External coverage/tau metrics are near-perfect under conservative calibration settings; this improves validity margin but can reduce sharpness/decision aggressiveness. - Checkpoint-level external policy sweeps (alpha/lambda/margin) did not find an alert-free setting that preserved acceptable quality simultaneously, so the conservative external configuration is retained. - Over-conservative readiness penalties are now audit-conditioned: penalty is applied only when all external datasets are near-perfect and all audited p-value profiles are strongly high. Current canonical external audits do not meet that stronger condition. - External FEMTO/XJTU/C-MAPSS artifacts now include backfilled `audit_*.json` and `audit_cache_*.npz` diagnostics from the saved external checkpoints.

1.11 10. Auto-Compiled Figures (Canonical Build)

This section is auto-generated from canonical artifacts and is intended for release packaging.

1.11.1 External Metrics Table

Dataset	RMSE	MAE	RMSE_last	MAE_last	RUL_cov	Tau_v
femto	14.362	7.993	52.873	50.395	1.000	0.000

Dataset	RMSE	MAE	RMSE_last	MAE_last	RUL_cov	Tau_v
xjtu_sy	30.133	23.684	33.171	33.156	1.000	0.000
cmapss	16.396	10.992	16.623	12.219	1.000	0.000

1.11.2 Figures

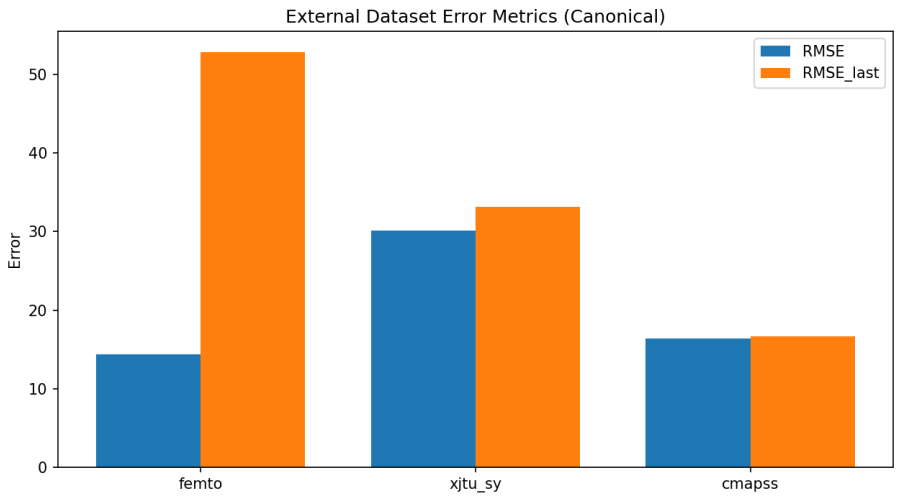

Figure 1: External Metrics Overview

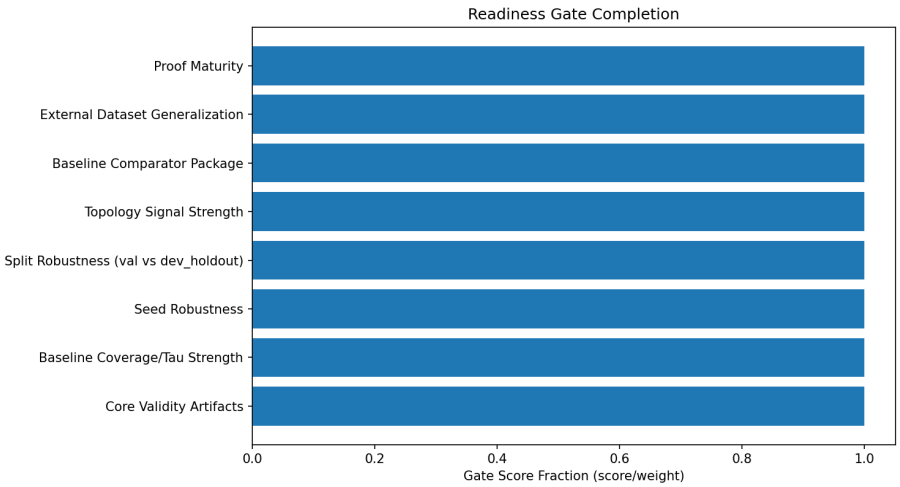

Figure 2: Readiness Gate Completion

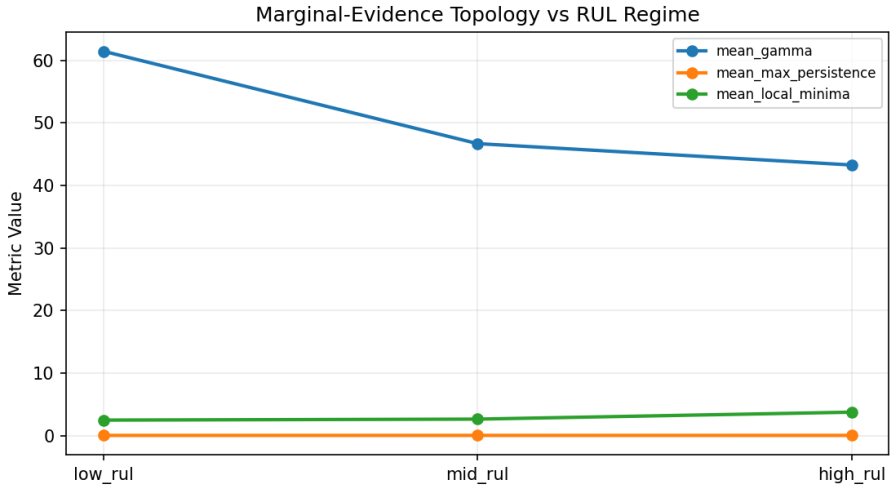

Figure 3: Topology vs RUL Bins

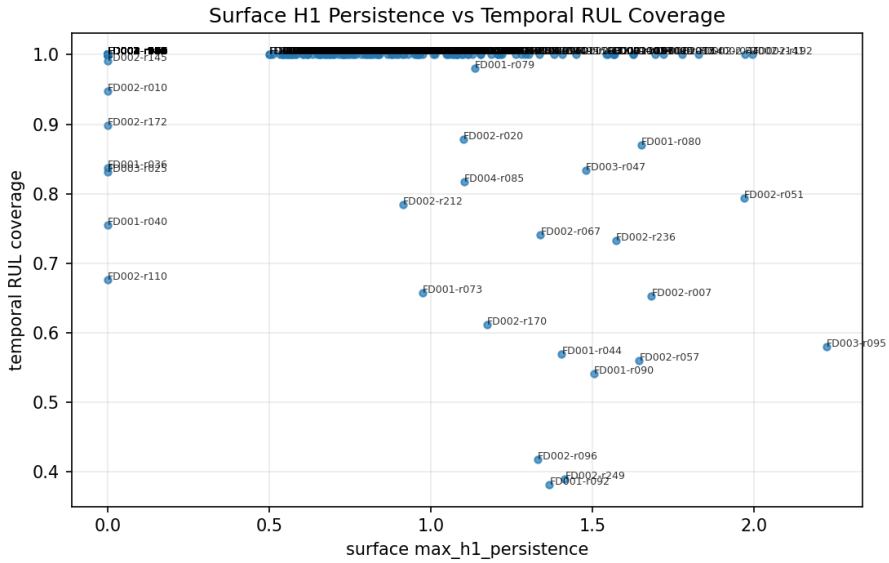

Figure 4: Surface H1 vs RUL Coverage

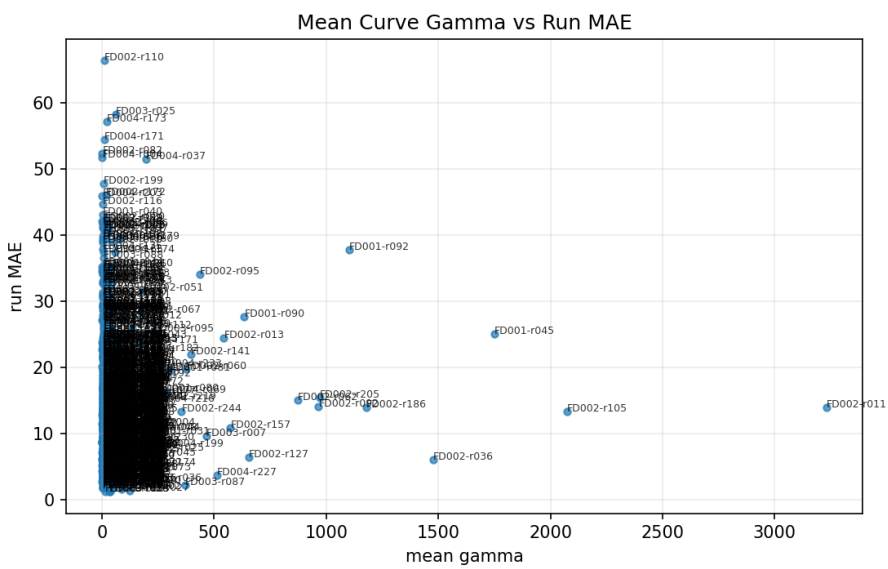

Figure 5: Gamma vs Prediction MAE